Mihai-Cristian **Farcaș**

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EDUCATION

- **Babeș-Bolyai University** **Cluj-Napoca, Romania**
  MSc in Cloud, Network Infrastructure and High-Performance Computation *2026 - Present*
- **Babeș-Bolyai University** **Cluj-Napoca, Romania**
  BSc in Computer Science (English), Faculty of Mathematics and Computer Science *Oct 2023 - June 2026*
  - **Thesis**: Hybrid post-quantum key exchange for WireGuard, the first FIPS 203-compliant hybrid WireGuard implementation on a production Rust codebase.

EXPERIENCE

- **BluePlan** **Software Engineer (Freelance)**
  Atlas: RAG Platform *2026 - Present*
  - Built and shipped Atlas end to end: retrieval-augmented chat over technical documents with hybrid dense, sparse and image retrieval fused with RRF, reranking and verbatim citations pinned to the source PDF. Multi-tenant, multilingual and in daily production use. Python, FastAPI, Qdrant, React.
- **Baseline Studios** **Full-Stack Developer**
  Client Applications *2026*
  - Delivered full-stack features across multiple client applications (stratosphere.vip, pursuitofx.org, kabu.xyz, esoftplus-landing.vercel.app) using TypeScript and React.
  - **Technologies**: TypeScript, React.
- **GeoDraw** **Software Developer**
  AI-Powered Design Canvas *2025*
  - Developed an AI-powered canvas for sketching and wireframing apps with React, TLDraw and LLM APIs, turning natural language into accurate real-time design output on the canvas.
  - **Technologies**: React, TLDraw, LLM APIs.
- **Tvarita** **Software Development Intern**
  Website Redesign *April - June 2025*
  - Rebuilt and redesigned a server monitoring site for an optician clinic chain using React and Tailwind CSS, improving the user experience and significantly reducing load times.
  - **Technologies**: React, Tailwind CSS.

PERSONAL PROJECTS

- **Helio** **2026**
  - Open-source remote desktop and game streaming over a post-quantum WireGuard tunnel: one native app acts as client and host on macOS, Windows and Linux, with ML-KEM-768 encryption, NAT traversal through the BitTorrent DHT, Reed-Solomon loss recovery and HDR video at up to 120 fps.
  - **Technologies**: Rust, GPUI, ML-KEM-768, WireGuard, AV1/HEVC.
- **Atlas** **2026**
  - AI assistant for technical documents (PDF, Word, presentations and e-books) that answers in the user's language with every claim tied to a citation opening the exact highlighted passage; hybrid dense, sparse and image retrieval with RRF fusion and reranking, multi-tenant and in daily production.
  - **Technologies**: Python, FastAPI, Qdrant, React, LLM APIs.
- **asides** **2026**
  - Screen-annotation tool for reporting bugs out loud: freeze any app, pin what is wrong and say why; local Whisper transcription types the aside into a coding agent's chat as a message with the annotated screenshot and measured geometry.
  - **Technologies**: Rust, GPUI, Whisper, MCP.

- **pq-boringtun** **2026**
  - Hybrid post-quantum key exchange for WireGuard: ML-KEM-768 alongside X25519 in the Noise IK handshake, built in Rust on Cloudflare's BoringTun and benchmarked on Apple M4 and Raspberry Pi 5.
  - **Technologies**: Rust, ML-KEM-768, X25519, Noise Protocol, WireGuard.
- **Supernova** **2026**
  - Live-wallpaper engine for macOS in Swift 6: renders looping video and sandboxed web bundles behind desktop icons, with audio, cursor and motion reactivity, per-display assignment and smart playlists.
  - **Technologies**: Swift 6, Metal, Core Audio.
- **eBPF Network Analysis System** **2025**
  - Network traffic classification for QUIC protocols, inspecting encrypted flows at the kernel boundary using eBPF with C, Rust and Python.
  - **Technologies**: C, Rust, Python, eBPF.
- **nvim-search-and-replace** **2025**
  - Project-wide search and replace plugin for Neovim written in Lua: async streaming ripgrep search, live before/after preview, selective replacement and its own undo/redo stack.
  - **Technologies**: Lua, Neovim, ripgrep.

TECHNICAL SKILLS

Programming Languages: Rust, Go, Python, Swift, C/C++, TypeScript/JavaScript, Lua, Shell.

Frontend: React, Next.js, Astro, Tailwind CSS, TLDraw, GPUI, Vercel, Cloudflare Workers.

Backend & Data: FastAPI, PostgreSQL, Qdrant, FAISS, LangChain, PyTorch, LLM APIs.

AI & Retrieval: RAG, hybrid dense/sparse/image retrieval, RRF fusion, reranking, citation grounding, Whisper, MCP.

Systems, Networking & Security: eBPF, WireGuard, Noise Protocol, ML-KEM-768, post-quantum cryptography, NAT traversal, BitTorrent DHT, Reed-Solomon FEC, Metal, Core Audio.

Tools & Platforms: Git, Neovim, ripgrep, Linux, macOS, Windows.